

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question

$$y = x + 9$$

$$y = x^2 + 16x + 63$$

A solution to the given system of equations is (x, y) . What is the greatest possible value of x ?

Answer

- A. -6
- B. 7
- C. 9
- D. 63

Correct Answer: A

Rationale

Choice A is correct. It's given that $y = x + 9$ and $y = x^2 + 16x + 63$; therefore, it follows that $x + 9 = x^2 + 16x + 63$. This equation can be rewritten as $x + 9 = (x + 9)(x + 7)$. Subtracting $(x + 9)$ from both sides of this equation yields $0 = (x + 9)(x + 7) - (x + 9)$. This equation can be rewritten as $0 = (x + 9)((x + 7) - 1)$, or $0 = (x + 9)(x + 6)$. By the zero product property, $x + 9 = 0$ or $x + 6 = 0$. Subtracting 9 from both sides of the equation $x + 9 = 0$ yields $x = -9$. Subtracting 6 from both sides of the equation $x + 6 = 0$ yields $x = -6$. Therefore, the given system of equations has solutions, (x, y) , that occur when $x = -9$ and $x = -6$. Since -6 is greater than -9 , the greatest possible value of x is -6 .

Choice B is incorrect. This is the negative of the greatest possible value of x when $y = 0$ for the second equation in the given system of equations.

Choice C is incorrect. This is the value of y when $x = 0$ for the first equation in the given system of equations.

Choice D is incorrect. This is the value of y when $x = 0$ for the second equation in the given system of equations.